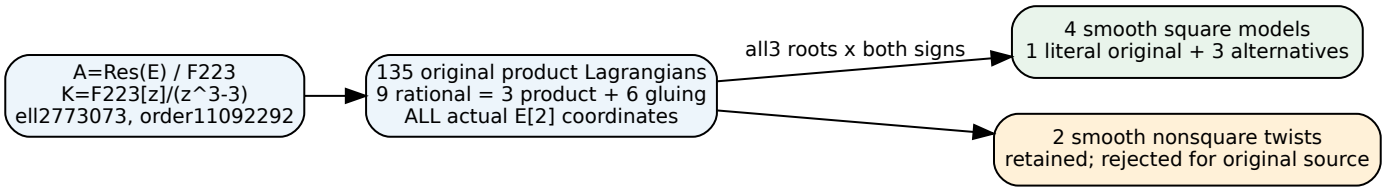
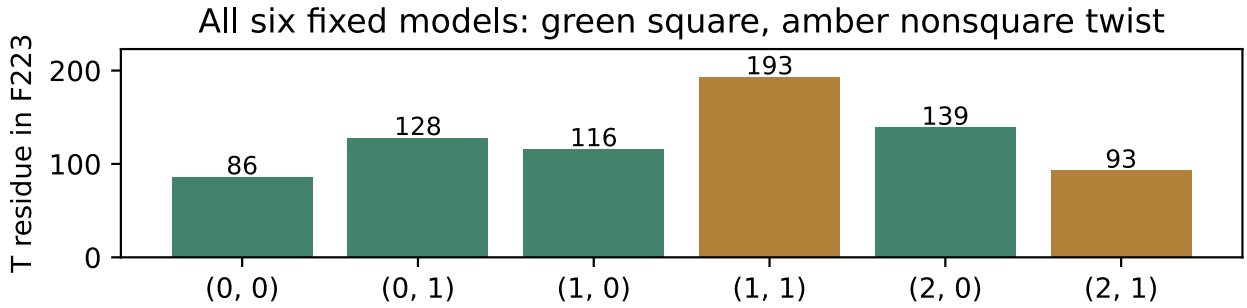
# Six classical models; four square candidates

2026-11-13 · actual N4 q223 / seed1 · unsigned / partial formalization




Fixed q223 source: K=F223[z]/(z^3−3), prime ell=2773073, #E(K)=11092292. All standing unsigned. Prospective27369d precedes model code; HLP arXiv:math/9809210v1 §4.1 Lemma13/Propositions14–15 was read and its complete PDF digest retained. This is a classical gluing construction, not a novelty claim.

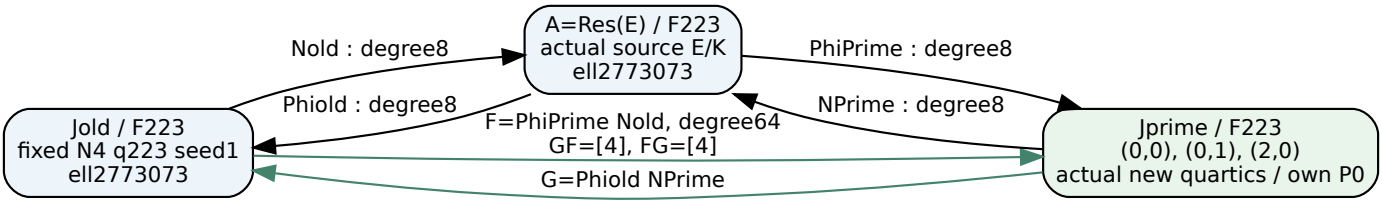
Native f33c42 audits all135 original Lagrangians, all4096 basis pairings and actual source2-torsion coordinates. Cyclic factor Frobenius fixes exactly9 kernels:3 product and6 distinct HLP gluings. Whole stable sets, rather than only counts, match the explicit kernels. The original81split/54nonsplit and identity/transvection controls remain.

For all3 source2-torsion roots and both signs, form H diagonal hi/off-diagonal−Ai with h2h3=Delta1 cyclically, Q=adj(H). Native checks verify the original HLP model after scaling by−1/4, twisting factor T=det(H), all15 Moore-descended coefficients, original smoothness and all three quotient Jacobians AX=D*Ai,BX=D²*Bi,D=16T. All six models are smooth; four are square eligible and two retain nonsquare twist obstructions.

(0, 0): T=86, D=38, square; (0, 1): T=128, D=41, square; (1, 0): T=116, D=72, square; (1, 1): T=193, D=189, nonsquare; (2, 0): T=139, D=217, square; (2, 1): T=93, D=150, nonsquare. Candidate(1,0) reproduces the original normalized15 coefficients exactly; the other arrays alone are not endpoint evidence.

Every model has224 rational curve points. Both Fp6 square-root signs satisfy the quotient equations and differ by actual2-torsion translation. Subtracting the actual P0 image gives sign-independent K descent. Compiler failure01/1e3436 is retained; the fix changes only calls to the original field constructor API.

# Actual whole-group maps and longer chains




Protocol626d84 fixes ALL four eligible models before map code. The new adapter recomputes quartic/P0/model tags, degree6 normal forms and raw trace origin. PhiPrime traces complete degree2 fibre ideals with multiplicity and subtracts only the actual Q=0 fibre. NPrime reconstructs the original length3 divisor algebra, sums all raw geometric images with multiplicity and subtracts the actual residual-line Dinf offset. It is a whole-group norm, not doubled-norm halving.

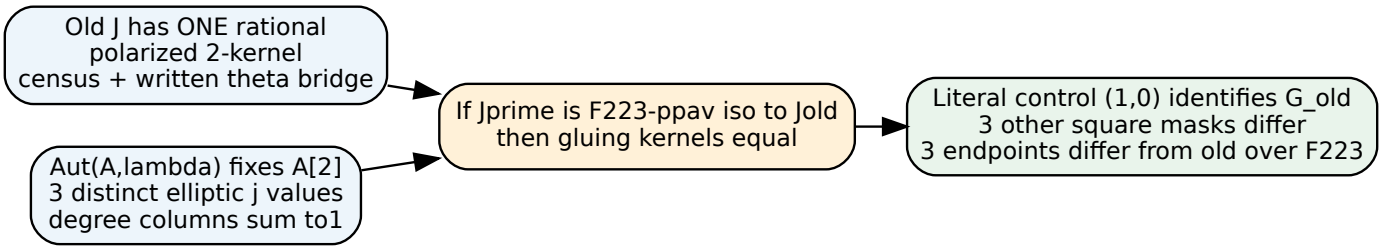
Native03/7c9bf4 passes one unfiltered test/all2401 input hashes unchanged. Every candidate checks32 source points, all224 new point classes and224 original point classes, both[2] compositions and both[4] chained compositions, nonzero exact-ell generator images, all five public targets and independent known-answer equalities. Runtime inputs contain coordinates and model tags only.

Each candidate checks100 pullback and100 norm addition pairs,48 alternate charts and all224 equal geometric line offsets. All12 sign-branch points carry dual-number doubled-fibre controls that reject the wrong linear fibre; multiplicity3 omission changes every candidate norm. Raw quotient infinity signs occur2+2 geometrically. Reachable K inverse-infinity counts are2/0/2/0; in the other cases alpha is nonsquare in K, so those infinities are not K-rational. Fresh alternate-P0 tags, zero origins and off-curve controls pass.

First map native01/ec6f14 compiled but failed the trace-origin rank check because the fresh adapter used the default cubic basis. The corrected adapter pins x^3−3 and compares all9 basis products and3 Frobenius images to independent FpR. Whole01, intermediate pass02/8e56fa and expanded03 are preserved; no fitted shift/scalar, source selection or original engine change.

Exact target identifiers for the three alternatives in the diagram: (0, 0) = Ciani-Jac-gluing-v1-a8cc8ba660e95d3f46749163261f5a110205b9098a1479f8147b77dd2eabcb20; (0, 1) = Ciani-Jac-gluing-v1-ac87c38dbd2dd728f73fc15b5ce27d6f1956e44687c8f6cfc214fc78ca2b6330; (2, 0) = Ciani-Jac-gluing-v1-d01edb6c77cc16a9a954d328ba858930939749cd2cbe9d5c0681b2d7d2c1d14b. identities.json gives every full normalized equation and P0, including the original control.

# Why the three alternatives differ from old J




Written5f7e4c: the global Ciani V4 quotient is genus0; its group-algebra trace vanishes on the Jacobian. The three degree2 quotient pullback/norm pairs therefore give both[2] compositions; cross homomorphisms vanish by the induced involution−1 and connected image in finite2-torsion. Frobenius permutes the factors, square D lies in F223 and auxiliary sqrt(alpha) choices change only translations, so the induced degree-zero maps descend.

Thus F=PhiPrime*Nold and G=Phiold*NPrime satisfy GF=[4],FG=[4]. Odd prime ell makes4 invertible, preserving the original DLP. The literal original candidate returns the old map with fixed sign+1 and the trivial[2] chain. This positive control is retained.

Endpoint argument: the old J has one rational polarized2-kernel. Any F223-ppav isomorphism from a candidate to old J would identify their dual quotient maps up to a polarized automorphism of A. The three elliptic j conjugates are distinct and not0/1728. Restricting the product polarization to each factor forces a single degree1 component, so every such automorphism acts by factor signs and fixes A[2]. Hence its gluing kernel must equal the original literal-control kernel. The other three square kernel masks differ, so each target differs from old J over F223. Mutual or geometric-extension isomorphism classes are not asserted.

All four rational cofactor groups are C4: that attempted endpoint discriminator is negative. Eleven feasible Lean lemmas pass both unchanged4.35/4.34.1 gates with zero trusted declarations:8 general model identities and3 chain/injectivity/degree implications. Exact local PCC bindings are unsigned. Polynomial/degree lemmas use standard propext/Classical.choice/Quot.sound; two functional lemmas use no axioms. Actual abelian/Picard/V4/Galois/theta/polarization geometry remains written, not fully formalized.

No computational advantage or security conclusion follows. Complete map counters, original S3 checkpoint/live budget/capacity admission and strongest reference algorithms remain open. Next: clean independent full/own CI, genuine whole-data handoff, then actual chain-kernel structure and further nondual walks. Existing V2/bitangent/kernel reports remain frozen; this new graph adds the actual alternative route with its unsigned and partial-formal scope.